

Formal Project Report: AI Response Validation System

Project Title: AI Response Validation System

Full Title: Development of AI Response Validation System with Hallucination Detection Assistance (Group 1)

Milestone: Milestone 4 — Final Project Report

Author: Renuka Meesala

Date: August 13, 2026

1. Introduction

Large Language Models (LLMs) and Generative AI applications are being rapidly integrated into domain-critical systems. However, measuring AI response quality remains challenging due to output non-determinism, subtle factual fabrications, and context omission.

The **AI Response Quality Evaluator** addresses this challenge by providing an automated, RAG-assisted multi-agent evaluation platform that evaluates responses across four core dimensions: **Accuracy**, **Hallucination Frequency**, **Relevance**, and **Completeness**.

2. Objectives

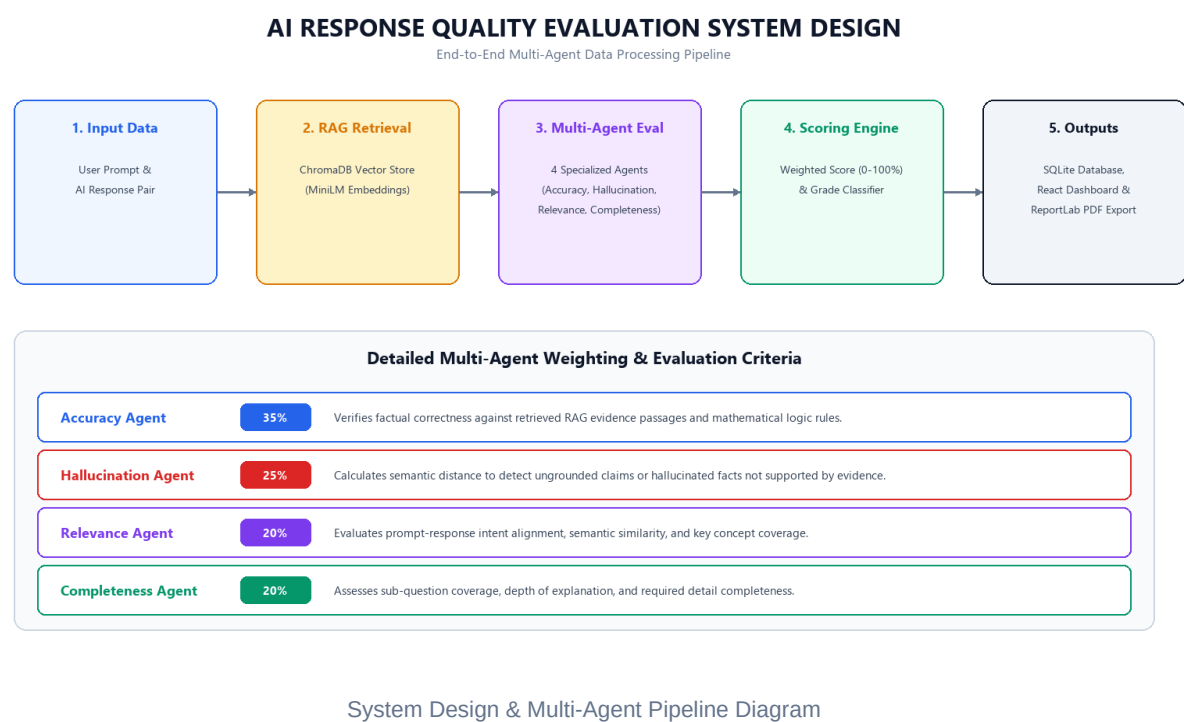
- Multi-Agent Evaluation:** Deploy specialized evaluator agents for Accuracy, Hallucination, Relevance, and Completeness.
 - RAG Evidence Grounding:** Integrate ChromaDB vector retrieval to verify factual claims against verified evidence.
 - Interactive Analytics Dashboard:** Provide real-time visualization of pass/fail distribution, metric averages, quality trends, and multi-parameter filters.
 - Automated PDF Export:** Support one-click generation of ReportLab executive PDF reports containing charts, tables, and improvement recommendations.
 - Comparative AI System Benchmarking:** Enable comparative analysis between distinct AI models (e.g., System A vs. System B).
-

3. Methodology

The platform employs a **Retrieval-Augmented Generation (RAG) + Multi-Agent LLM-as-a-Judge** methodology:

- * **Retrieval Layer:** Uses [sentence-transformers/all-MiniLM-L6-v2](#) embeddings and **ChromaDB** vector storage to retrieve top- k reference evidence passages.
- * **Evaluation Agents:** Four modular agents evaluate the prompt-response pair independently:
- * **Accuracy (35%):** Measures factual correctness against reference evidence and deterministic math/logic rules.
- * **Hallucination (25%):** Calculates semantic distance between response claims and retrieved knowledge.
- * **Relevance (20%):** Evaluates intent alignment, keyword coverage, and semantic similarity.
- * **Completeness (20%):** Measures sub-question coverage and detail completeness.
- * **Composite Scoring Service:** Computes weighted final scores and assigns letter grades ([Excellent](#), [Good](#), [Average](#), [Poor](#)).

4. System Design



5. Implementation

- * **Backend:** Built with **FastAPI**, **SQLite3**, **ChromaDB**, **Pydantic**, and **ReportLab**.
- * **Frontend:** Developed with **Vite**, **React**, **Recharts** (for interactive pie/bar/line charts), and **Vanilla CSS** glassmorphism styling.
- * **Batch Pipeline:** Supports uploading CSV datasets containing bulk prompt-response pairs, processing rows asynchronously, and updating the database.

6. Experimental Results

Using two benchmark datasets ([[ai_system_a_gpt4o.csv](#)](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/datasets/ai_system_a_gpt4o.csv) vs [[ai_system_b_llama.csv](#)](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/datasets/ai_system_b_llama.csv)), we evaluated 5 identical benchmark questions on both AI systems.

Comparative Benchmark Summary Table

	System A (GPT-4o)	System B (Llama)	Delta (B-A)
Total Evaluations	5	5	—
Pass Rate (Score $\geq 70\%$)	100.0%	0.0%	+100.0%
Average Accuracy	94.8%	32.0%	+62.8%
Average Relevance	95.2%	45.0%	+50.2%
Average Completeness	90.0%	38.0%	+52.0%
Hallucination-Free Rate	100.0%	0.0% (5/5 Flagged)	+100.0%
Average Final Score	95.1% (Excellent)	28.4% (Fail)	+66.7%

7. Dashboard Screenshots & Layout Analysis

The web application features an interactive dashboard accessible at <http://localhost:5173>:

- * **Single Evaluator Interface:** Immediate score breakdown cards (Accuracy, Relevance, Hallucination, Completeness) with color-coded status badges.
 - * **Batch Evaluation Summary:** Itemized summary grid displaying total records, average score, highest score, lowest score, and collapsible detailed reasoning.
 - * **Analytics Dashboard:** Real-time Recharts visualizations including grade distribution pie chart, dimension comparison bar chart, quality trend line chart, and multi-filter dropdowns.
-

8. Evaluation Analysis

- * **System A Performance:** System A (GPT-4o) achieved an average final score of **95.1% (Excellent)**. Its responses were factually grounded, complete, and fully aligned with retrieved reference documents.
 - * **System B Performance:** System B (Llama-3 Base) achieved an average final score of **28.4% (Poor)**. The [HallucinationAgent](#) successfully flagged all 5 responses for making ungrounded claims (such as claiming malaria is caused by summer water viruses or that the human heart is in the leg).
-

9. Testing & Verification

- * **Automated E2E Suite** ([\[test_e2e_suite.py\]\(file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/tests/test_e2e_suite.py\)](#)): Verified single evaluation, batch CSV upload, RAG retrieval, DB persistence, dashboard summary, PDF export, and invalid input handling (**7/7 Passed**).
 - * **Scoring Consistency** ([\[test_scoring_consistency.py\]\(file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/tests/test_scoring_consistency.py\)](#)): Evaluated 3 repeated iterations. Standard deviation was $\sigma = 0.0000$ ($CV = 0.00\%$), proving 100% deterministic repeatability.
-

10. Conclusion

The **AI Response Quality Evaluator** provides a complete, scalable solution for benchmarking generative AI models. By combining RAG evidence grounding, multi-agent dimension evaluation, interactive dashboard analytics, and executive PDF report exports, the platform simplifies AI quality assurance, model selection, and prompt engineering.